Biological and climatic constraints drive reproductive synchrony
Supplementary figures

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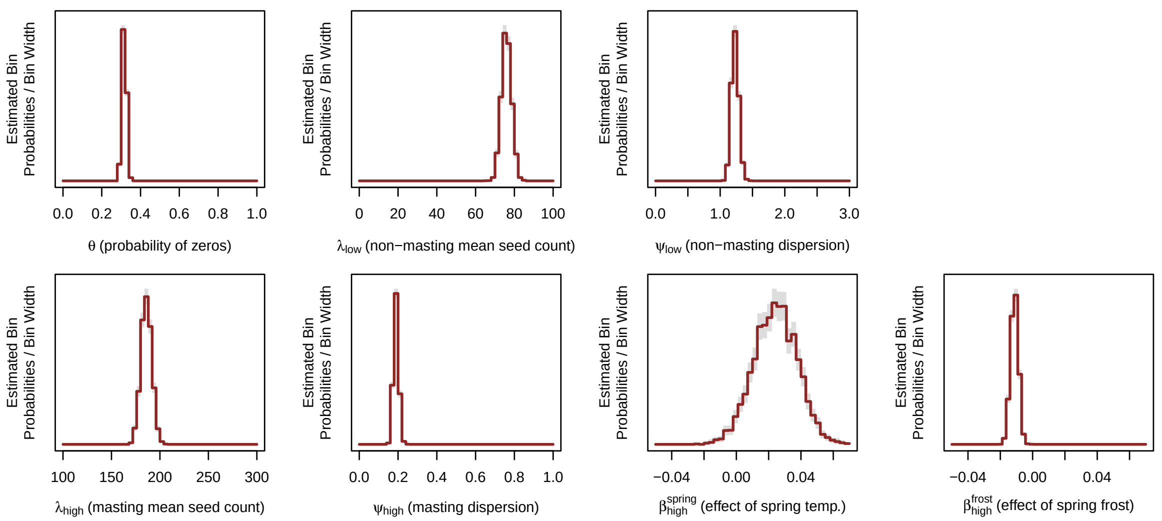

Figure 1: **Posterior distributions of the parameters related to seed count distributions (main text model).** These parameters control the number of seeds produced in low and high states. *Spring temp.* corresponds to the average spring temperature in April and May. *Spring frost* correspond to the growing degree days accumulated until the last frost day. See Methods for details.

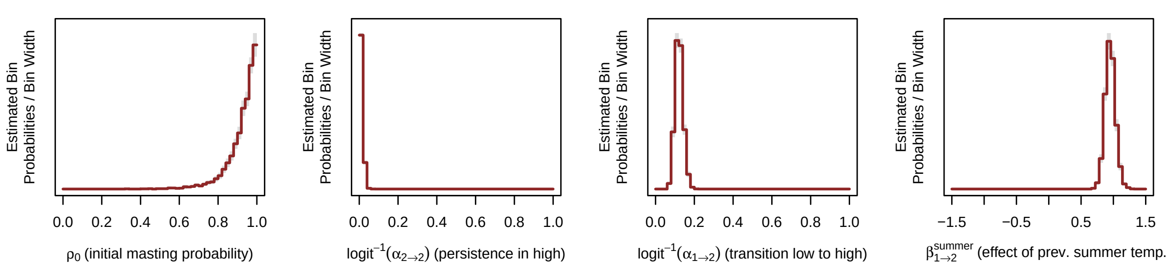

Figure 2: **Posterior distributions of the parameters related to state transitions (main text model).** These parameters control the transition probabilities between states (1 = low, 2 = high). *Prev. summer temp.* corresponds to the average maximum temperature in July and August (of the previous year). See Methods for details.

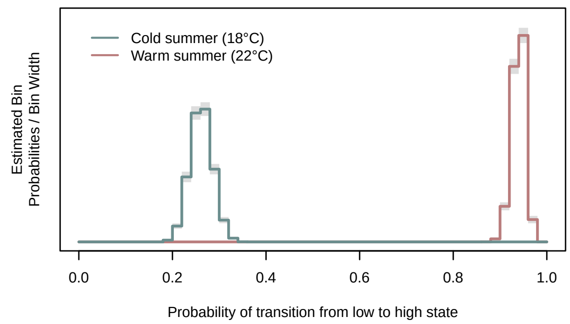

Figure 3: **Posterior distributions of transition probability from low to high reproductive states in a cold vs. a warm summer.** On average, the transition probability during a warm summer is 3.6 times higher than during a cold summer.

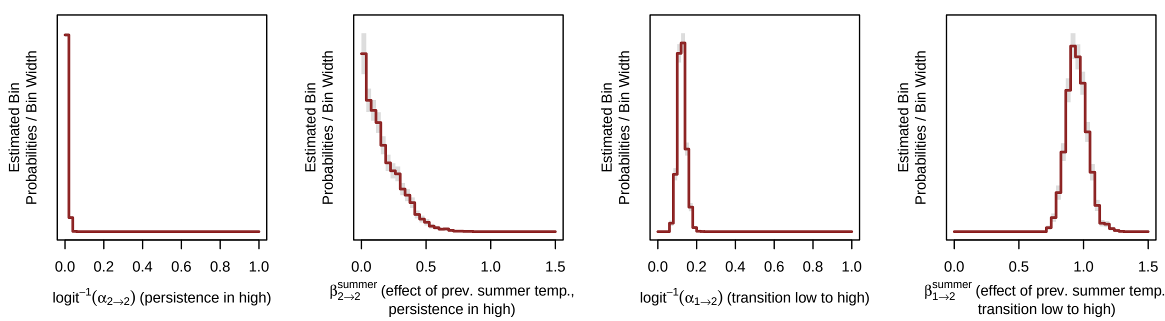

Figure 4: **Posterior distributions of the parameters related to state transitions (model with summer temp. on both transitions).** In this model, we tested whether the persistence in a high state was affected by previous summer temperature ($\beta^{\mathsf{summer}}_{2\to2}$). Because this model is more complex (one additional parameter), we constrained both beta parameters to be > 0, otherwise the model would not fit. This lower bound constraint was not present in the model presented in the main text.

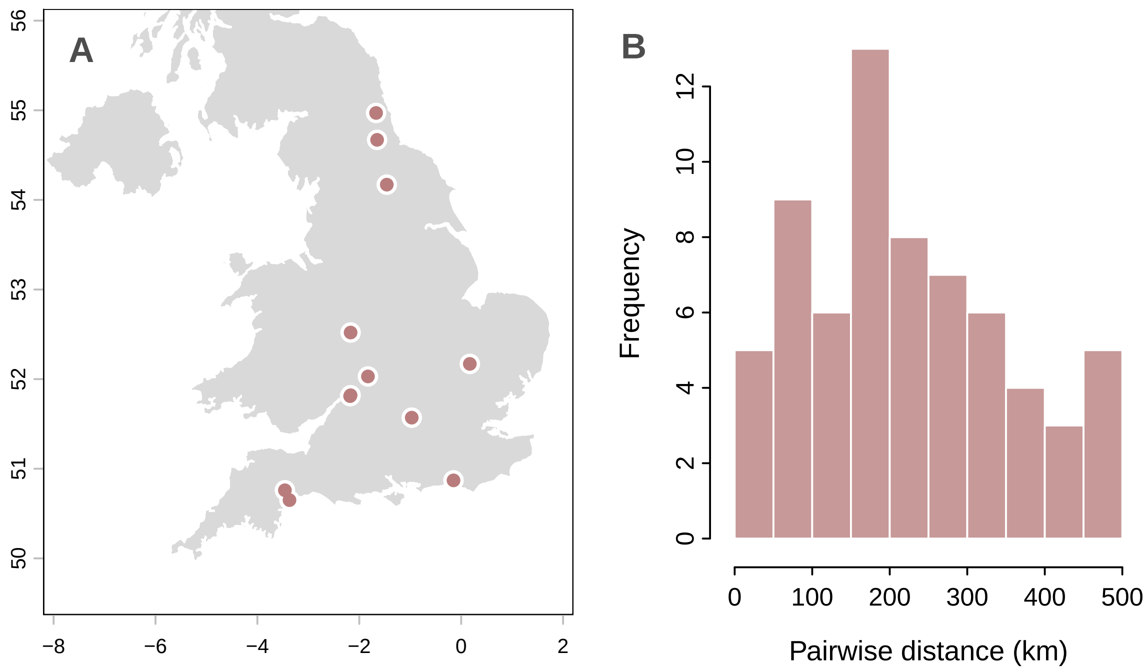

Figure 5: **(A) Map of the beech forest sites (England).** We used data from 171 trees on 12 sites. **(B) Distributions of pairwise distance between sites.** Sites are on average 224 km apart.

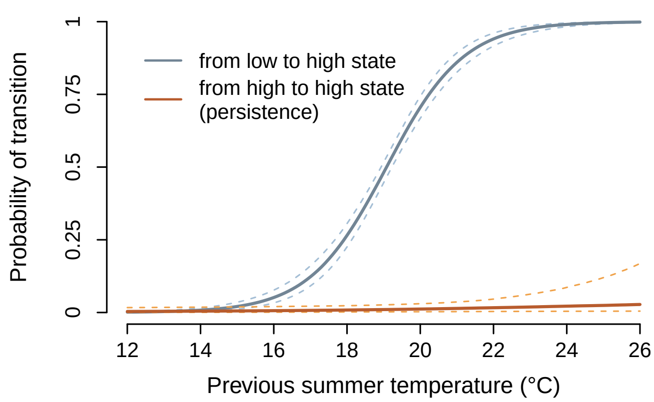

Figure 6: **Effect on previous summer temperature on the probability of transition from a low reproductive to a high reproductive state, and on the probability of persistence in a high reproductive state (model with summer temp. on both transitions, see Supp. Fig 4).** Even when summer temperature could influence both the transition to a high state and the persistence in a high state, trees could not become stuck indefinitely in a high state.